Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Auto-elevating and expanding canopy system for temperature stabilization in building construction.*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI** (√), Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√)**, Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical 、 Materials		*Mechanical 、 Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Mechanical Automation Technologies*	*Robotics/Automation Machinery*	*Measurement/Sensor Technologies for Automation*
Concept of Proposed Technology	The research aims to develop an automated modular top-covering system with automated elevation for building construction sites that provides environmental protection and thermal control to support stable concrete curing. The system is installed at the top of the structure and automatically elevates as construction progresses, maintaining continuous coverage of the active work zone. By integrating self-climbing, deployable canopy mechanisms, and adaptive fabric handling, the system enables weather-resilient, temperature-controlled construction operations. The system reduces costs and safety risks associated with manual canopy reinstallation [1–3].			

	Key features: - Automatic lifting mechanism synchronized with floor-level construction progress (e.g., high-rise apartment construction) [3,4,5] - Automated expanding and folding canopy structure for dynamic coverage adjustment - Adaptable grasping and tensioning mechanisms for stable canopy deployment - Modular system design for flexible implementation across building geometries
Rationale for Joint R&D Project	The proposed project requires close collaboration between an academic research group and external partners to translate a laboratory-scale prototype into a deployable construction system. Differences between controlled laboratory conditions and the complex weather environments of the Korean peninsula necessitate a joint R&D approach. While the RICAL group has strong expertise in system architecture, control design, and prototyping, it lacks field-level resources for real-world construction validation, particularly for thermal and environmental performance. The proposed system is primarily intended for standardized building types, such as high-rise apartment construction commonly found in Korea. In addition, the dynamic and seasonal characteristics of Korean weather, including prolonged winter conditions, provide an appropriate and demanding testbed for a weather-responsive automated canopy system. Effective deployment in these environments requires domain-specific construction knowledge and on-site validation capabilities that extend beyond in-house laboratory resources. **Rationale for Collaboration:** - The RICAL group contributes expertise in automated structural systems and prototype design. - Partner institutions contribute advanced material selections and a construction domain connection for Proof of Concept on a real construction site - Joint system-level design and validation are necessary to meet performance and deployment requirements **Expected Benefits of Joint R&D:** - Improved thermal and environmental performance through advanced automation system integration - Reduced technical risk through collaborative design and field-oriented validation - Accelerated transition from laboratory prototype to site-deployable construction technology
Key Words	1. Automation in Construction 2. Automatic Origami Scaffolding System 3. Modular Components 4. Construction Safety and Workers' Health 5. Adaptive Flexible Fabric Control

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	The objective of the joint R&D project is to develop an advanced automatic canopy system for building construction sites that can vertically elevate in synchronization with floor-level progression throughout the construction process. The system incorporates an origami-inspired structural scaffold [8] that enables automated expansion and folding, allowing the canopy to dynamically adapt to changing construction stages and site conditions. By eliminating the need for manual canopy reinstallation at each floor rise, the proposed system significantly reduces repetitive labor, minimizes work at height, and improves operational efficiency. Furthermore, the automated canopy provides a controlled internal environment that supports stable concrete curing by mitigating the effects of external weather conditions. This environmental control also enhances worker safety, comfort, and productivity, contributing to more reliable and resilient construction operations. The specific R&D objectives are: - To develop an automatic elevation system that reliably elevates the canopy assembly in synchronization with floor-level construction progress. - To develop an automatic expanding and folding canopy mechanism capable of supporting multiple covering materials (e.g., fabric, polymer sheets, and carbon-based composites) and adaptively enclosing the active construction area. - To develop an AI-driven smart canopy control system that determines immediate expansion and folding strategies based on environmental sensor inputs, including humidity, temperature, and ultraviolet (UV) intensity [7], to maintain optimal conditions for concrete curing and worker safety. - To develop a modular system architecture that allows flexible deployment across diverse building geometries and construction layouts while reducing manufacturing, installation, and operational costs. - To design and integrate a robust control and safety management framework that ensures stable operation under repeated elevation, expansion, and folding cycles in variable environmental and load conditions [3,5]. - To validate the performance, reliability, and scalability of the integrated system through laboratory testing and real-world construction site demonstrations.			

R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Temperature Control within Coverage Area (85% ≥ area in ±2°C band):** The system shall maintain a curing-supporting temperature band (setpoint ±2–3°C) across at least 85% of the covered work area for at least 80% of the operating time. The target temperature range will be set to support concrete curing requirements while maintaining a comfortable working environment for onsite personnel. Performance will be evaluated based on spatial temperature uniformity and the sustained stability of the temperature band throughout representative work durations. **Intelligent Environmental Sensing and Adaptive Coverage Control (Decision Accuracy ≥ 90%):** The system shall employ AI-driven sensing and decision-making algorithms to determine optimal canopy coverage based on multi-modal environmental sensor inputs, including temperature, humidity, UV intensity, and other relevant jobsite conditions. The canopy shall autonomously adapt its expansion and folding behavior in response to changing environmental states to maintain curing-supporting conditions and improve worker safety. Performance will be evaluated using Event Detection Accuracy (≥90%). The system shall detect atmospheric events such as rain onset, excessive UV exposure, high wind conditions, and curing-risk temperature/humidity bands with an event-level detection score of F1 ≥ 0.90. **Weather-Delay Time Reduction (30% ≥):** The system shall reduce weather-related work stoppages by at least 30%, thereby contributing to an overall construction schedule reduction of 10% through improved work continuity and more consistent curing conditions. This target is motivated by evidence that U.S. construction projects can lose approximately 45–60 workdays per year due to adverse weather conditions [6,7]. While these statistics are reported for U.S. construction, similar or greater weather-related productivity losses are observed in Korea due to prolonged winter conditions and seasonal weather variability. Performance will be evaluated by comparing weather-induced downtime and schedule delays under conventional operations versus canopy-assisted operations, including delays attributable to work interruption and extended curing requirements. **Cost Reduction (Equipment Idle Time Reduction ≥ 10%, Financial reduction ≥ 5%):** Construction cost reduction is expected through multiple mechanisms enabled by the automated canopy system. First, schedule compression will reduce financing costs (e.g., accrued interest and capital engagement duration) by at least 5%. Second, by mitigating weather-related interruptions and improving curing consistency, the system will reduce equipment idle time and extended rental periods by at least 10%. Additional cost benefits are expected from reduced labor demand for repetitive and high-risk canopy installation tasks, including decreased overtime and improved productivity. Performance will be assessed through comparative cost breakdown analyses based

	on labor hours, equipment utilization logs, and time-dependent indirect costs. **System Reliability and Operational Safety (Availability ≥ 97%, MTTR ≤ 30 min):** The system shall achieve an operational availability of at least 97% during active construction periods, with a mean time to repair (MTTR) of ≤30 minutes for recoverable failures. Reliability performance will be evaluated using failure frequency, downtime duration, and recovery time across repeated elevation, expansion, and folding cycles. Operational safety will be verified through controlled testing and field demonstrations under varying environmental and construction conditions, ensuring stable operation without hazardous failure modes.
R&D Plan	This R&D project will be carried out over three years using a phased development and validation approach that advances from simulation and concept-based studies to laboratory testing and ultimately to real-world construction environments. Plans for each year are listed below. ● **Year 1**: Knowledge-based Structure Design and Simulation Analysis (TRL 2, 3) ○ Investigate the current KPI structure of apartment construction projects in Korea and its climate environment ○ Design an automatic canopy system ■ Define conceptual layouts of the canopy, supporting structure, and interface with building cores or formwork systems ○ Design an automatic self-elevation system ■ Design mechanical lifting and climbing mechanisms synchronized with floor-level construction progress ■ Evaluate alternative elevation concepts through kinematic and structural simulations ○ Develop a control algorithm for automatic canopy and elevation systems ■ Define control architecture linking elevation, expansion, and folding subsystems ■ Develop preliminary control logic for synchronized operation and fail-safe behavior ■ Conduct structural and thermal simulations to verify feasibility under representative loads and environmental conditions ○ Define system performance metrics and evaluation protocols for temperature control, reliability, and safety ● **Year 2**: Laboratory-Scale Prototyping and Real-World Preparation (TRL 3, 4, 5) ○ Small-scale prototype development and feasibility test ■ Fabricate a functional laboratory-scale prototype incorporating elevation, canopy deployment, and sensor systems ■ Validate basic functionality and operational feasibility under controlled conditions ○ Optimize control algorithms and mechanical structures for stable elevation,

expansion, folding, and synchronized operation.

- Conduct laboratory experiments to evaluate robustness, safety, and task success rates
  - Conduct accelerated lifecycle and durability testing for repeated elevation and folding cycles
  - Perform repeated operation cycles to assess mechanical durability and control stability
  - Evaluate system response to simulated environmental variations such as temperature, humidity, and wind
- Develop a user-friendly user interface and control device
  - Design an intuitive user interface for system monitoring, manual override, and safety control
- Design a real-world-sized device based on laboratory feasibility test
  - Translate laboratory findings into a full-scale system design suitable for construction site deployment
- Preparation and scaling up from laboratory-scale to real-world implementation
  - Develop installation procedures, manuals, and maintenance guidelines
  - Conduct pre-deployment safety checks and compliance reviews

- **Year 3**: Real-World Deployment and Technology Advancement (TRL 5, 6)
  - Deploy a large-scale automatic canopy system in a real construction site environment
  - Conduct long-term field operations to evaluate system reliability, safety, and environmental performance
    - Validate AI-driven sensing and adaptive control performance under complex and dynamic site conditions
  - Validate temperature control performance and construction time reduction under real weather conditions
    - Quantitatively assess reductions in weather-induced delays and curing time compared to conventional construction practices
  - Assess labor, cost, and equipment efficiency improvements compared to traditional practices
    - Analyze a break-even point in labor hours related to canopy installation
    - Evaluate improvements in equipment utilization rates and reductions in idle time and rental periods
    - Estimate overall construction cost savings, including indirect costs such as financing and schedule-related expenses
  - Refine system design based on field feedback and operational data
    - Identify performance limitations, failure modes, and usability issues observed during field deployment
    - Implement targeted design refinements to improve reliability and

	operational efficiency ○ Establish guidelines for standardization and future commercialization pathways

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for developing the core system intelligence, control algorithms, and performance evaluation frameworks for the proposed automated canopy system. The institute will lead the foundational research on automated elevation and deployment logic, as well as sensor-driven decision-making methods, to enable reliable, adaptive operation under dynamic construction and environmental conditions. The specific roles and responsibilities of the research institute include: ● Development of control algorithms for automatic elevation, expansion, and folding of the canopy system in synchronization with floor-level construction progress ● Research on AI-driven environmental sensing and adaptive decision-making using temperature, humidity, and UV sensors to optimize canopy coverage for concrete curing and worker safety ● Design of overall system architecture, including control logic, data flow, and modular software-hardware integration frameworks ● Simulation-based development, laboratory validation, and quantitative performance evaluation of automated canopy behaviors under varying environmental and operational scenarios ● System-level integration of mechanical structures, actuators, sensors, and embedded control hardware ● Analysis of experimental data to identify performance gaps and guide system optimization ● Provision of technical guidance and close collaboration with the SME partner to support system integration, testing, and iterative refinement Through these activities, the research institute will ensure the proposed system delivers robust intelligence, reliable control performance, and measurable improvements over conventional construction-site canopy solutions.
Expected R&R of Korean SME Partner	The SME partner will be responsible for the design, manufacturing, integration, and validation of the physical canopy system and associated hardware components. The SME will play a critical role in translating research outcomes into a practical, scalable, and deployable construction technology suitable for real-world site conditions. The specific roles and responsibilities of the Korean SME partner include:

	• Feedback on the mechanical design and manufacturing of the automatic elevation system and origami-inspired expanding and folding canopy structure in real-world scale • Selection, testing, and integration of suitable covering materials and structural components for durability, thermal performance, and safety in construction environments • Execution of real-world construction site experiments, including identifying test sites, system installation, operation, and performance validation • Collection and analysis of field data to evaluate system reliability, safety, and environmental control performance • Contribution to system robustness improvement, cost reduction strategies, and commercialization planning based on manufacturing expertise and field deployment experience • Development of deployment guidelines, safety protocols, and standardization strategies for application under Korean construction regulations Through these responsibilities, the SME partner will ensure the proposed system is manufacturable, field-ready, and has a clear pathway to commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcome of this project is a fully functional automated canopy system that can be reliably deployed on real building construction sites and that elevates in synchronization with floor-level progression. The project will deliver an integrated and validated prototype comprising an automatic elevation mechanism, an expanding and folding canopy structure, and AI-based environmental sensing and control modules. System performance will be verified through staged laboratory experiments and full-scale field demonstrations, where technical feasibility, operational reliability, and safety will be evaluated under representative construction workflows and environmental conditions. In terms of operational outcomes, the proposed technology is expected to improve construction productivity and strengthen worker safety. By automating repetitive canopy installation and elevation operations, the system will reduce labor demand and mitigate schedule delays caused by weather-dependent stoppages. In addition, by reducing worker exposure to adverse weather and to high-risk operations at height, the system will help lower accident and near-miss rates. Stable environmental control within the covered workspace is also expected to improve concrete curing and workability, supporting improved quality assurance, reduced rework, and more predictable project schedules. From an industry and societal perspective, this project will advance smart, automated construction technologies aligned with national construction digitalization initiatives. The resulting system will provide a scalable and modular solution applicable to standardized building projects, particularly

	high-rise apartment construction, where repetitive floor-by-floor work cycles are common. Furthermore, the research institute and SME collaboration framework will accelerate technology transfer, strengthen domestic industrial competitiveness, and establish a commercialization pathway, enabling wider adoption of intelligent construction site protection systems and supporting the long-term modernization of the construction sector.

6. References

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